Alexander DeLise

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Education

Florida State University May 2027
B.S. Applied and Computational Mathematics, B.S. Computational Science, GPA - 4.0 *Tallahassee, FL*

- **Relevant Coursework:** Computational Probabilistic Modeling, Data Mining and Machine Learning, MCMC Methods, Discrete/Continuous Algorithms for Science, Programming for Science, Math Modeling, Statistics
- **Honors in the Major Thesis:** Topic – Machine Learning and Data Science, Advisor – Dr. Nick Dexter, WIP

Experience

Emory University June 2025 – August 2025
Research Fellow — Scientific Machine Learning and Data Science | *Atlanta, GA*

- Developed Bayesian methods for optimal rank-constrained encoder–decoder neural networks, connecting to dimensionality reduction and inverse problems
- Implemented and trained autoencoders in PyTorch on biomedical imaging and financial datasets, benchmarking against theoretically optimal mappings
- Performed large-scale data preprocessing, feature extraction, and model evaluation (MSE, RMSE, explained variance), identifying significant reconstruction gaps in standard training approaches, and even deep neural networks
- Collaborated in a diverse team, honing communication, adaptability, and joint problem-solving to drive research progress
- Co-authored a manuscript on Bayesian machine learning and neural networks, under submission to a peer-reviewed journal (preprint on Google Scholar)

University of Tennessee, Knoxville May 2024 – April 2025
Research Fellow – Quantum Algorithms | *Knoxville, TN*

- Built preprocessing pipeline to identify and prepare feasible solutions for constrained binary optimization problems
- Developed a quantum-state-labeling method with ancillary variables, achieving up to 360x improvement in solution quality compared to baseline heuristics
- Implemented and benchmarked quantum optimization algorithms with custom state preparation using Python/Pennylane
- Designed a reproducible evaluation suite by serializing parameter configurations into callable files for scalable experiments
- Coordinated within a cross-disciplinary group to troubleshoot, share insights, and reach common goals for steady progress
- Co-authored a manuscript demonstrating method under submission to a peer-reviewed journal (preprint on Google Scholar)

FAMU-FSU College of Engineering September 2023 – June 2025
Undergraduate Researcher – Operations Research | *Tallahassee, FL*

- Designed and implemented large-scale optimization models for pandemic-era patient allocation and healthcare facility planning across Florida's 67 counties
- Processed and analyzed extensive COVID-19 datasets (hospitalization, vaccination, geospatial) and integrated them with epidemic SIRV dynamics and policy constraints into a decision-making framework
- Simulated over 1 million patient transfers, achieving a 16.4% reduction in unmet hospital demand and identifying bottlenecks in urban healthcare networks
- Developed pipelines in Python, R, CPLEX, and GAMS for data preprocessing, model implementation, and geospatial visualization of demand and capacity imbalances
- First author on a paper accepted to the 2025 North American IEOM Conference and presented findings at the 2025 Florida Undergraduate Research Conference (DOI forthcoming)

Projects

Image-to-LaTeX Generation with Deep Learning | *Python, PyTorch, SQL, Albumentations, SentencePiece, Kaggle* |

- Designed an end-to-end pipeline (SQL + PyTorch) to convert mathematical expression images into LaTeX
- Implemented data ingestion with SQL for CSV preprocessing with 100k+ samples from the Im2Latex-100k dataset
- Developed CNN–Transformer architecture with 2D positional encodings, label smoothing, and beam search
- Automated training and evaluation with SentencePiece tokenization, mixed precision, and checkpointing
- Delivered full prediction pipeline with single-image inference, beam-search decoding, and LaTeX rendering

Quadratic Portfolio Optimization Using Gurobi | *Python, Gurobi, NumPy, pandas, yFinance* |

- Built a Mixed-Integer Quadratic Programming model to optimize portfolios with real-world investment constraints
- Processed 14 years of price data for 250+ stocks; computed returns and covariance matrix
- Achieved portfolios with 35%+ annualized returns while limiting sector exposure and annualized risk below 30%
- Automated result reporting with selected allocations, expected return, portfolio risk, and sector breakdown

Technical Skills

Languages: Python, R, MATLAB, C++, SQL, LaTeX
Tools: Git, GitHub, VS Code, IBM ILOG CPLEX, GAMS, Gurobi, Microsoft Office Suite
Frameworks and Libraries: NumPy, PyTorch, pandas, SciPy, scikit-learn, gurobipy, TensorFlow, Pennylane, tidyverse, ggplot
Involvements: *Participant*, JPMorganChase Data for Good Hackathon | *President* Pi Mu Epsilon Math Honor Society